

Evaluating the Robustness of NLP Models for Phishing Email Detection

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Abstract

This project evaluates the robustness of NLP-based phishing detection models under text obfuscation and their reliance on keyword-based features. We compare seven models, including Gaussian Naive Bayes, TF-IDF Multinomial Naive Bayes, TF-IDF Logistic Regression, CNN, BiLSTM, and transformer-based models (BERT, DistilBERT), trained on the Nazario and Enron datasets using text and engineered features.

All models are evaluated on clean and obfuscated data using a homograph attack. Results show high recall across models, with transformers achieving the best overall performance while remaining robust to obfuscation. Explainability analysis reveals consistent reliance on phishing-related keywords.

1 Introduction

Phishing emails remain one of the most prevalent and damaging vectors of cyberattack, exploiting human psychology rather than technical vulnerabilities. Many security systems rely on NLP-based models to automatically detect them. However, attackers can use text obfuscation techniques that may bypass detection, particularly if models rely on specific keywords rather than contextual understanding.

This project investigates two related hypotheses: (1) common text obfuscation techniques significantly reduce the performance of NLP-based phishing detection models (defined as a drop greater than 10% in recall), and (2) these models rely heavily on a small set of phishing-related keywords when making predictions.

To test these hypotheses, we evaluate seven models spanning a range of architectural complexity: Gaussian Naive Bayes, TF-IDF Multinomial Naive Bayes, TF-IDF with Logistic Regression, a character-level CNN, a BiLSTM with pre-trained embeddings, and transformer-based models (BERT and DistilBERT). The dataset combines two widely used benchmarks in phishing detection: the Nazario Phishing Corpus (Nazario, 2007) and the Enron Email Dataset (Klimt and Yang, 2004).

Each model is evaluated on both clean and obfuscated versions of the test set, where obfuscation is introduced through homograph substitution (Dalmiere et al., 2025). We further apply explainability analysis using LIME and gradient-based saliency to identify which words drive phishing predictions.

2 Related work

Early approaches to phishing detection relied on traditional classifiers, using email features such as number of URLs, presence of JavaScript, and domain age as indicators of phishing (Fette et al., 2007). The field has since evolved to include machine learning and deep learning approaches, from Naive Bayes to transformer-based models (MohamedAli and Abduhameed, 2024). However, these models are often trained on datasets that exclude obfuscated emails, leading to reduced performance against real-world phishing techniques (Dalmiere et al., 2025). Furthermore, model explainability is rarely analyzed in prior work.

One of the key limitations in phishing email detection is the lack of rich, diverse, and well-curated datasets. In prior work, multiple datasets from different sources were compiled for machine learning-based phishing detection (Champa et al., 2024). Although these datasets are large and balanced, the definition of "phishing" often differs between sources and is not explicitly documented.

Phishing is defined as "the fraudulent attempt to obtain sensitive information by disguising oneself as a trustworthy entity in digital communication" (Desolda et al., 2021). In some curated datasets, emails labeled as phishing are in fact unsolicited messages that neither impersonate a trusted sender nor attempt to extract sensitive information. As a result, models may learn patterns that do not reflect true phishing behavior.

Another dataset initially considered was the University of Twente Phishing Validation Emails Dataset (Miltchev et al., 2024), due to its recency and alignment with our definition of phishing. However, manual inspection revealed a large number of duplicate entries and limited linguistic di-

versity, as the dataset was generated from a small set of templates. This would likely lead models to recognize fixed phrases rather than generalize to real phishing patterns.

For these reasons, we use the Nazario Phishing Corpus (Nazario, 2007) and the Enron Email Dataset (Klimt and Yang, 2004). The Nazario dataset contains real-world phishing emails collected from active campaigns, while the Enron corpus provides a large collection of legitimate corporate communications. This combination allows for a clear binary labeling scheme and is widely used in phishing detection research.

3 Method

3.1 Data Collection

The Nazario and Enron datasets are first parsed into a unified format, where each email is assigned a binary label (phishing = 1, legitimate = 0). After filtering invalid or low-quality samples (e.g., empty or malformed messages), the datasets are merged and split into training, validation, and test sets using a stratified 80/10/10 ratio to preserve class distribution. To ensure a controlled evaluation, each split is further balanced by downsampling the majority class. Preprocessing includes extracting the subject and body, cleaning HTML content, normalizing text, and masking obvious patterns such as URLs, email addresses, and numerical tokens to reduce overfitting to surface-level artifacts.

3.2 Feature Engineering

In addition to raw text, we extract a set of lightweight structural and lexical features that capture properties commonly associated with phishing emails. These include measures of message length, counts of URLs and domains, presence of suspicious HTML elements (e.g., forms or scripts), and indicators of abnormal formatting or urgency. In total, we construct 15 engineered features that complement textual representations. This design follows prior work showing that combining text with structural features improves phishing detection, by capturing behavioral patterns not obvious from lexical content alone (Fette et al., 2007).

3.3 Experimental Design

We evaluate seven different models to compare their behavior in phishing detection. For text-based models, the text is defined as the concatenation of the email subject and body. For each model,

hyperparameters and preprocessing decisions (e.g., use of stop words) are tuned on the validation set, selecting the configuration with the highest recall, given that a false negative can be more costly than a false positive in this context. Then, the models are trained on the training set and evaluated on the test set using precision, recall, F1-score, accuracy, and confusion matrix.

To validate the first hypothesis, we conduct a stress test by applying obfuscation techniques to the phishing emails on the test set and comparing recall, F1, and accuracy against the original. The method to obfuscate the test set is a rule-based homograph attack targeting phishing-related keywords based on prior work (Dalmiere et al., 2025). Only phishing emails are modified, where selected terms (e.g., login, password, verify, urgent, bank) are detected using case-insensitive matching with word boundaries. Each matched keyword is perturbed using a character-level substitution, replacing Latin characters with visually similar Cyrillic homographs (e.g., $a \rightarrow \text{a}$, $e \rightarrow \text{e}$). The transformation is applied to both email subject and body fields when available, generating an obfuscated test set for robustness evaluation. This leaves legitimate emails untouched so that we can measure the degradation in model performance. While this approach provides a controlled setting for evaluation, it does not reflect more complex real-world obfuscation strategies (which we discuss in Section 5).

To validate the second hypothesis, we applied explainability techniques to identify the top 10 most frequent words driving predictions on the test set emails, and the percentage of emails in which each word appeared as a top feature. Different explainability methods were used depending on model type and input representation. Classical text models (TF-IDF NB, TF-IDF LR) and sequence/transformer models (BiLSTM, BERT, DistilBERT) used LIME because they operate on word-level inputs without inherent interpretability, while the CNN used gradient-based saliency since it processes characters and allows direct gradient attribution to the input.

3.4 Models

Gaussian Naive Bayes The Gaussian Naive Bayes receives the 15 engineered structural features as input, without any text, for comparison. Hyperparameters are tuned on the validation set, using grid search over `var_smoothing=0.1`, selecting the one with the highest recall. The best

configuration is trained on the training set. Due to the fact that it does not use the text as input, SHAP is used to explain feature contribution as a complementary result (not reported here, as it is not directly related to the hypotheses), and it is immune to obfuscation.

TF-IDF Naive Bayes Multinomial Naive Bayes with Term Frequency-Inverse Document Frequency (TF-IDF) using text only as input. The hyperparameters smoothing $\alpha=0.5$, vocabulary size=50000, n-gram range=(1,1), and minimum document frequency=1 are tuned using random search (Bergstra and Bengio, 2012) on the validation set, and the best configuration is selected based on the highest recall. Aggregated LIME was applied across all of the test emails, and, finally, the model was evaluated on the obfuscated test set.

TF-IDF Logistic Regression The model combines TF-IDF vectorization with Logistic Regression (LR). The experiment was conducted in two stages: evaluation of three input configurations (text-only, features-only, and text+features). The winning architecture from Stage 1 was carried forward to Stage 2, which performed a grid search over n-gram range (unigrams, bigrams, trigrams), vocabulary size (10k, 50k, 100k features), regularisation strength C (0.01, 0.1, 1.0, 10.0), and sublinear TF scaling. Model selection at both stages was based on the highest validation recall; false negatives were prioritized because missed phishing emails can be costly. The best LR configuration uses unigrams (n-gram_range=(1,1)), max_features=10000, $C=10$, and sublinear_tf=True on plain text only. LIME was used to analyze which words drive phishing predictions, and the model was evaluated on the obfuscated test set.

Character-level CNN The architecture uses a character embedding layer, parallel convolutional filters with max-over-time pooling, and a fully connected output layer, which enables the model to learn character-level patterns without requiring tokenization. Hyperparameters and preprocessing decisions were made on the validation set with a defined seed for reproducibility, using the highest recall as the decision metric. The hyperparameters were tuned using random search (Bergstra and Bengio, 2012) over the number of filters, kernel sizes, dropout, learning rate, and maximum

sequence length; while the preprocessing decisions concerned whether to use text only or text combined with features.

The best configuration (max_chars=1024, num_filters=512, kernel_sizes=[5,7,9], dropout=0.5, lr=1e-3, input as text only) was trained on the training set, and the model checkpoint with the highest recall was saved to a pickle file. To test the second hypothesis, gradient-based saliency (Simonyan et al., 2013) was applied to identify the top 10 most frequent words driving the predicted labels across all test emails. Finally, the model was evaluated on the obfuscated test set.

BiLSTM with pretrained embeddings The model is a two-layer Bidirectional Long Short-Term Memory network (BiLSTM) with GloVe pre-trained word embeddings (GloVe 6B, 300 dimensions), with out-of-vocabulary tokens initialized uniformly at random. The vocabulary was built from the training set with a minimum frequency threshold of 2. The architecture consists of a two-layer bidirectional LSTM followed by a classifier head ($\text{hidden_dim} \times 2 \rightarrow 128 \rightarrow 2$). When using features, they are passed through a separate MLP ($15 \rightarrow 32$) and concatenated with the LSTM output before the classifier. This allows structural email signals to be taken into account. The same two-stage pipeline was applied. Training used AdamW with weight decay 10^{-4} , cross-entropy loss, a learning rate scheduler that reduces on plateau with factor 0.5 and patience 2, and early stopping with patience 3. Hyperparameters were fine-tuned for learning rate (1e-3, 5e-4, 1e-4) and hidden dimension (128, 256, 512) to select and save the configuration with the highest validation recall; it was also evaluated on the obfuscated test set. The best BiLSTM configuration uses lr=1e-4 and hidden_dim=256 on plain text only. LIME was applied across 30 phishing and 30 legitimate test emails to identify the top 10 most frequent words driving predictions.

Transformer-based Models We fine-tune pre-trained transformer models, namely BERT (Devlin et al., 2019) and DistilBERT (Sanh et al., 2019), for phishing email classification. The input is constructed by concatenating the email subject and body, and the contextual embedding of the [CLS] token is used as a text representation for classification.

In addition to text-only inputs, we explore a hy-

brid setup where engineered structural features are projected through a small feed-forward network and concatenated with the transformer representation. This design follows prior work showing that auxiliary features can improve performance when they provide complementary information (Huang et al., 2020). Feature engineering can introduce complementary structural signals (e.g., formatting patterns and metadata), but its impact on transformer-based models is minimal, with a performance improvement in accuracy and F1 of 0.02% for BERT. This is because transformer models already learn rich contextual and structural representations directly from raw text, capturing many of the same patterns targeted by engineered features. The best models, BERT hybrid and DistilBERT, are reported in Section 4.

Among the transformer variants, BERT (hybrid) and DistilBERT (text-only) are selected as the best models and are trained using cross-entropy loss with the AdamW optimizer, using a maximum sequence length of 256, batch size of 16, learning rate of 2×10^{-5} , and dropout of 0.3, with early stopping based on validation performance (patience = 2).

4 Results

Table 1 presents the performance of each model on the original test set. All models achieve high recall, with TF-IDF NB, TF-IDF LR, BiLSTM, and transformer-based models reaching 0.994 or higher. Gaussian NB achieves the lowest precision (0.591), as it relies only on structural features without text context.

Table 2 shows results on the obfuscated test set. TF-IDF LR shows the largest recall drop (6.94%), while CNN, TF-IDF NB, BiLSTM, and transformer-based models show marginal drops of under 1.72%. Gaussian NB is immune to text obfuscation as it uses only engineered structural features rather than text input. No model exceeds the 10% recall drop defined for the first hypothesis.

Table 1: Original Test Set Results

Model	Prec.	Rec.	F1	Acc.
Gaussian NB	0.591	0.934	0.724	0.644
TF-IDF NB	0.989	0.994	0.991	0.991
TF-IDF LR	0.989	0.994	0.991	0.991
CNN	0.977	0.989	0.983	0.983
BiLSTM	0.966	0.994	0.980	0.980
BERT (hybrid)	0.972	0.994	0.983	0.983
DistilBERT	0.972	1.000	0.986	0.986

Table 2: Obfuscated Test Set Results

Model	Prec.	Rec.	F1	Acc.	Rel. Recall Drop (%)
Gaussian NB	0.591	0.934	0.724	0.644	0.00
TF-IDF NB	0.988	0.983	0.986	0.986	1.11
TF-IDF LR	0.988	0.925	0.956	0.957	6.94
CNN	0.977	0.983	0.980	0.980	0.61
BiLSTM	0.971	0.983	0.977	0.977	1.11
BERT (hybrid)	0.971	0.983	0.977	0.977	1.16
DistilBERT	0.971	0.983	0.977	0.977	1.72

To validate the second hypothesis, explainability techniques are applied across all test emails. In Table 3, we report the top 3 of the 10 most frequent words (due to limited report space) driving phishing predictions per text-based model (therefore, it does not include Gaussian NB), along with the percentage of phishing-predicted emails in which each word appeared as a top feature. Note that percentages are not directly comparable across models, as different explainability methods were used (LIME for text-based models, and gradient-based saliency for CNN). Words such as "account", "bank", and "your" appear as top features across different models, indicating the presence of finance-related terms in phishing predictions.

Model	Top-3 Features
TF-IDF Naive Bayes	account (20.1%) your (16.7%) bank (12.6%)
TF-IDF Logistic Regression	bank (60.0%) account (50.0%) member (33.3%)
Character-level CNN	account (75.0%) bank (51.7%) your (50.6%)
BiLSTM (pretrained)	bank (33.3%) account (30.0%) customer (26.5%)
BERT	ebay (40.0%) account (40.0%) paypal (30.0%)
DistilBERT	account (50.0%) your (50.0%) ebay (40.0%)

Table 3: Top contributing features across models

5 Discussion and conclusion

The first hypothesis (text obfuscation significantly reduces model performance by more than 10% in recall) is not supported. Despite targeting phishing-related keywords with homograph substitution, no model degraded more than 7% in recall. TF-IDF

LR has the largest drop (6.94%). The other models have only marginal drops, generally below 1.72%.

The structural differences between models can explain their varying robustness to obfuscation. TF-IDF-based models treat obfuscated words as out-of-vocabulary tokens, ignoring them while relying on the remaining signals. However, TF-IDF LR showed a larger drop than TF-IDF NB (6.94% vs 1.11%), suggesting that logistic regression is more sensitive to vocabulary changes than the probabilistic NB model. BiLSTM uses pretrained embeddings and maps unknown tokens to generic vectors, but retains enough contextual signal from surrounding words to make the predictions. The character-level CNN processes characters, so it sees the substituted characters. However, its small performance drop (0.61%) suggests that it learned enough character patterns that changing a few characters was not enough to substantially degrade performance. Transformer-based models (BERT, DistilBERT) use contextual embeddings, which help maintain performance under obfuscation (1.16% and 1.72% recall drop). This shows that models using contextual or character-level information are less sensitive to obfuscation than those relying on exact token matches. These results suggest that models rely on multiple complementary signals, allowing them to remain robust to partial keyword-level obfuscation.

The obfuscation technique applied provides a controlled evaluation of lexical perturbations but does not capture more complex real-world text obfuscation techniques, which may lead to an overestimation of model robustness. The second hypothesis (models rely on a small set of phishing-related keywords) is supported across all models. Financial and brand-related terms such as "account", "bank", and "PayPal" consistently appear as top predictive features, with some terms appearing as a top feature in over 75% of phishing-predicted emails, suggesting that keyword reliance is a general property of NLP-based phishing detection. For legitimate emails, "meeting", "fw:" and "attached" appear as top predictive features in multiple models, reflecting the corporate vocabulary of the Enron dataset. This suggests that models may rely on lexical cues rather than semantic understanding.

Overall, more robust detectors should move beyond keyword-based representations toward contextual understanding, for example, through adversarial training or more diverse datasets that reduce reliance on dataset-specific vocabulary.

6 Limitations

The use of older datasets (Klimt and Yang, 2004) may not reflect current real-world scenarios for deployment. In addition, it may make the classification task artificially easy due to the distinctive nature of corporate versus phishing emails, and it does not include current phishing techniques (e.g., SMS).

The obfuscation strategy is another limitation. It targets a set of predefined keywords, leaving emails that do not contain those terms unmodified. Therefore, the results may overestimate the models' robustness, as the strategy may not be reflected in real-world text obfuscation scenarios.

Future work should address these limitations by evaluating models against stronger obfuscation strategies targeting full email content, using more recent and diverse datasets, and adversarial training approaches that approximate real-world attacks.

Statement of contributions

Daria Goptsii handled data preprocessing, feature engineering, and transformer models (BERT, DistilBERT). Giane Mayumi Galhard implemented Gaussian NB, TF-IDF NB, and the CNN. Sarah Al Taleb implemented TF-IDF Logistic Regression, BiLSTM, and the obfuscation method. All authors contributed to experimental design, analysis, and report writing.

Use of generative AI technologies

In accordance with the course policy, generative AI tools were used only in an assistive manner. Grammarly was used to check for grammar and style errors. ChatGPT was used to format Python text output.

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